

VIPASHA SHARMA

Geospatial Analyst | GIS & Remote Sensing Specialist | Environmental Resource Mapping & Monitoring | One Health

CONTACT

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EDUCATION

PhD — Geomatics Engg.
IIT Roorkee · PMRF Scholar
2020–2026 | CGPA 9.75

M.Tech — Environmental Engg.
JUIT Waknaghat
2015–2017 | CGPA 8.1

B.Tech — Civil Engg.
MM University Mullana
2011–2015 | CGPA 8.63

GIS PLATFORMS

ArcGIS / ArcGIS Pro
QGIS
Google Earth Engine

REMOTE SENSING

Multispectral Analysis
Image Classification
ERDAS Imagine / ENVI
SAR Processing

PROGRAMMING & DATA

Python (GeoPandas, ArcPy)
SQL / Spatial Queries
ML

SPATIAL ANALYSIS

LULC & Change Detection
Risk & Suitability Analysis
Network / Proximity Anal.
Geodatabase Management

AWARDS & HONOURS

🏆 Prime Minister's Research Fellowship (PMRF) — 2021
🎓 UGC NET-JRF — 2019
✈️ GHS Travel Bursary — 2024
🌐 ISPRS TIF Grant — 2023

LANGUAGES

English — Professional
Hindi — Native

PROFESSIONAL SUMMARY

Geospatial Analyst and GIS & Remote Sensing Specialist with 8+ years of experience transforming satellite imagery, spatial datasets, and environmental data into high-impact research and decision-support tools. Core expertise spans KML boundary validation, LULC classification using multi-temporal satellite imagery, encroachment and proximity analysis, LULC change detection, geodatabase creation and management, drone data validation, and ML-based spatial modelling. Experienced with QGIS, ArcGIS, and Google Earth Engine. Published in ISPRS and IEEE. Fully set up for independent remote work.

EXPERIENCE

QGIS Expert / GIS Specialist | Mar 2026 – Present
Micro1 — AI Training & Data Platform | Remote / Global

- Lead end-to-end QGIS projects from raw spatial data acquisition and cleaning through to final map outputs, covering topology validation, KML boundary correction (overlaps, self-intersections), and deliverable production.
- Organise, clean, and validate diverse spatial datasets; manage projections, coordinate systems, and spatial references for optimal interoperability across parcel-level and boundary datasets.
- Troubleshoot complex QGIS geometry issues and document problem-solving strategies; build advanced cartographic layouts with documented design rationale.
- Contribute spatial QC workflows and reasoning to AI training data for next-generation geospatial analysis models.

PhD Research Fellow (PMRF) | Aug 2020 – Nov 2025
IIT Roorkee — Geomatics Engineering | Roorkee, India

- LULC mapping and change detection using multi-temporal Sentinel-2 and Landsat 5/7/8 imagery
- Vegetation analysis: NDVI, EVI, SAVI, NDWI on Sentinel-2 and MODIS time-series to study greenness dynamics and land-cover changes linked to environmental patterns.
- Created geodatabases — feature datasets, coordinate systems, domains, relationship classes, and topology rules — for large-scale spatial projects with ward-level accuracy.
- Developed ML environmental-health prediction models using Python with NASA POWER climate data; published in ISPRS Archives, ISPRS Annals, and IEEE Xplore.
- Canopy height model, glacial flow velocity, sediment transport analysis.
- Taught GIS and Remote Sensing to UG and PG students.

Assistant Professor | Jul 2017 – Jun 2020

Lovely Professional University — Civil & Environmental Eng. | Punjab, India

- Taught Environmental Engineering, EIA, Water Quality Management, and GIS fundamentals.
- Mentored student projects in watershed analysis, LULC mapping, vegetation analysis, and pollution monitoring.
- Introduced cartography and map design modules; contributed to Environmental and GIS lab development.

KEY PROJECTS

Geodatabase Creation, Data Cleaning & Spatial Querying

ArcGIS · QGIS · SQL · Python · GeoPandas

Created geodatabases with feature datasets, CRS, attribute domains, and topology rules. Performed vector QC and executed complex spatial queries for data extraction and multi-table joins directly applicable to ward-level shapefile and geodatabase.

Spatio-temporal cluster and hotspot analysis

ArcGIS · QGIS · SQL · Python · GeoPandas

Hotspot analysis, Kernel Density Estimation, Quadrat Analysis, Nearest Neighborhood analysis, Distribution analysis, Standard Deviation Ellipse, SATSCAN, Origin Destination Analysis, Service-Area/Location Allocation.

Multi-Factor Environmental Risk Assessment

Sentinel-1 · CartoDEM · ArcGIS · OSM

Composite risk index combining Sentinel-1 waterlogging, CartoDEM drainage, LULC change metrics, road proximity, and socio-demographic layers, MCDM, AHP, GWR

Spatio-Temporal LULC Change Detection

Sentinel-2 · Landsat 5/7/8 · ArcGIS · Google Earth Engine · R

Multi-temporal satellite imagery analysis, LULC classification, vegetation index time-series (NDVI, EVI, SAVI), and seasonal/ temporal change detection

Geo-spatial Dengue Risk Prediction Tool

Sentinel-2 · Landsat 8 · SRTM · Python · scikit-learn

End-to-end ML pipeline combining NDVI/EVI/LST covariates, LULC change metrics, DEM-derived waterlogging, and temporal dynamics for geospatial risk forecasting; included geotagging, incidence assessment, and field data validation workflows.

Road Network Accessibility & Service Area Analysis

ArcGIS Network Analyst · QGIS · OSM · Python

Developed road-network-based accessibility models for healthcare, emergency response, and public service coverage assessment. Performed shortest-path routing, service-area generation, travel-time estimation, and location-allocation analysis using OpenStreetMap road networks and network optimization techniques.

Flood Susceptibility & Waterlogging Mapping

Sentinel-1 · Sentinel-2 · CartoDEM · ArcGIS · Google Earth Engine

Generated flood susceptibility and urban waterlogging maps using SAR backscatter analysis, DEM-derived hydrological parameters, rainfall proxies, drainage density, and land cover information. Applied weighted overlay and spatial modeling approaches for flood-risk zonation and infrastructure vulnerability assessment.

Urban Growth & Built-Up Expansion Analysis

Landsat · Sentinel-2 · Google Earth Engine · ArcGIS

Performed urban expansion analysis using multi-temporal satellite imagery and built-up indices such as NDBI and UI. Assessed urban sprawl patterns, peri-urban growth, and land transformation trends through spatial-temporal change detection workflows.

PUBLICATIONS

- Sharma V., Ghosh SK, Khare S. Prediction of Transmission Rates of Dengue in NCT Delhi Using Machine Learning. IEEE CAI 2024.
- Sharma V., Ghosh SK, Khare S. Determination of Spatio-Temporal Transmission Patterns of Dengue: NCT Delhi Case Study. ISPRS Geospatial Week 2023.
- Sharma V., Ghosh SK, Khare S. A Proposed Framework for Surveillance of Dengue Disease and Prediction. ISPRS Archives 2023.
- Ghosh S., Sharma V. Tracking of Disease: A Review of Technology for Next Generation Healthcare. Deep Learning in IoT for Next Generation Healthcare. 2024.
- Sharma V., Ghosh SK. A Cloud-Based Thick Client Web Interface for Dengue Disease Monitoring and Management (D2M2). IGARSS 2025, Brisbane, Australia